SHoEbox: gradient horizon

Completed November 5, 2026. Matched complete-episode training from the regional-facts control checkpoint.

Result

Removing the 16-event gradient cut did not solve integrated memory or fact learning. The full-horizon endpoint preserves 100.00% composition, compared with 98.82% for the truncated arm. Shared-fact accuracy rises from 30.47% to 34.57%, while delayed correction falls from 39.26% to 29.69%.

Held-out task	Truncated	Full episode	Change
Composition, K=2	98.82%	100.00%	+1.18 pp
Recall, gap 16	39.26%	38.09%	-1.17 pp
Correction, gap 16	39.26%	29.69%	-9.57 pp
Shared fact	30.47%	34.57%	+4.10 pp
Left-private fact	26.76%	29.30%	+2.54 pp
Right-private fact	29.69%	27.54%	-2.15 pp
A-only shared	32.81%	34.77%	+1.95 pp
A-only left-private	35.94%	34.96%	-0.98 pp
B-only shared	26.17%	23.83%	-2.34 pp
C-only shared	26.17%	26.95%	+0.78 pp
C-only right-private	26.17%	25.98%	-0.20 pp

Both arms complete 32 optimizer updates, with 2,048 composition cases and 512 complete episodes of each stream task. The sole paired intervention is whether gradients cross the 16-event chunk boundary. Each held-out stream condition/query uses 512 fresh episodes; composition uses all 592 held-out cases. Predictions use the common 256-byte output head. Four-value chance accuracy is 25%.

Validation selects **full/best.pt at phase update 32**. Endpoint scores above are reported regardless of checkpoint selection. Test scores never select a checkpoint.

Validation and composition

Arm	Update	Composition	All composition gates	Weakest task
truncated	0	100.00%	pass	20.31%
truncated	16	99.62%	pass	21.88%
truncated	32	98.67%	pass	15.62%
full	0	100.00%	pass	20.31%
full	16	100.00%	pass	20.31%
full	32	100.00%	pass	23.44%

Selection ranks all capability gates, composition preservation, minimum task accuracy, macro accuracy and lower macro NLL, in that order. Composition preservation includes both local lookups and exact table retention, as well as the final answer. A high answer score alone does not pass all gates.

Composition control	Truncated	Full episode
One round	25.00%	25.00%
Two rounds	98.82%	100.00%
Four rounds	43.24%	40.54%
Exchange disabled	25.00%	25.00%
Changed table, original target	27.20%	27.03%
Changed table, changed target	98.99%	100.00%
State reset before query	22.80%	25.17%

Both arms preserve the seven-node graph, sparse expert banks, recurrent state format and common byte decoder. They inherit identical weights, optimizer moments and composition RNG. New stream episodes start from reset state with seed 20261109; partial parent stream states are intentionally discarded in both arms.

This pair also changes episode loss weighting and optimizer frequency relative to the preceding phase. Those changes are shared by both arms. The paired comparison isolates gradient detachment; comparisons to the earlier phase do not isolate it.

Delayed memory and write retention

Memory condition	Truncated	Full episode
recall, 16, carried	39.26%	38.09%
recall, 16, reset	24.80%	24.80%
correction, 16, carried	39.26%	29.69%
correction, 16, reset	24.61%	24.61%
withheld correction, 16, old value	35.55%	31.25%
recall, 32, carried	33.20%	30.08%
recall, 32, reset	24.80%	24.80%
correction, 32, carried	27.15%	28.32%
correction, 32, reset	24.61%	24.61%
withheld correction, 32, old value	33.59%	28.32%

Gap 16 is the training setting; gap 32 tests a longer delay. Query reset removes carried state immediately before the answer. Withheld corrections are scored against the old observed value here; raw records also include agreement with the unseen new value.

Local memory head	Truncated: write / query	Full: write / query
recall: P	17.19% / 29.69%	60.94% / 28.12%
recall: Q	48.44% / 35.94%	28.12% / 29.69%
correction: P	35.94% / 28.12%	37.50% / 29.69%
correction: Q	35.94% / 32.81%	26.56% / 31.25%

Local diagnostics use frozen training-only heads on 64 held-out episodes per task after selection. They measure whether regional publications contain readable values at the observed write and the final query. They do not supply inference answers.

Fact acquisition and verification

Local fact head	Truncated: write / query	Full: write / query
P: shared	31.25% / 31.25%	23.44% / 20.31%
Q: shared	28.12% / 31.25%	18.75% / 20.31%
P: left	37.50% / 26.56%	43.75% / 28.12%
Q: right	23.44% / 25.00%	26.56% / 32.81%

Training budget	Truncated	Full episode
Expert executions	6,561,792	6,561,792
External events	46,080	46,080
Observed source occurrences	68,536	68,536
Composition cases	2,048	2,048
Complete correction episodes	512	512
Complete fact episodes	512	512
Training time, minutes	28.67	28.66

All **360 regression tests** pass. Tests cover forward/state/cost equality, full-gradient equivalence to an unsliced episode in FP64, final-query credit reaching an earlier write only with the full horizon, loss-only labels and exact interrupted/resumed updates. FP64 avoids cancellation in the untrained fixture's large normalization gradients; training uses FP32.

The audit reconstructs each training batch from the generators, compares all data hashes and budgets, verifies parent weights and optimizer moments, and recomputes validation selection. It also verifies 76 held-out prediction groups and local diagnostic arithmetic. Exports remove all four auxiliary heads and reproduce checked inference logits and states exactly; the inference graph has 252,478 parameters.

Next development step

Focus on fact acquisition and selective publication inside the regional experts: teach a producer to recognize an observed write, preserve its payload while processing distractors, and replace it when a correction arrives. Test that learned write/retain decision with the existing reset, withheld-update and source-retention controls, while preserving successful two-round composition.

M7 remains open. The integrated milestone also requires multi-seed replication and the natural-text experiment.

Reproduce

Extract the source and artifact ZIPs into the same directory. Both use the SHoEbox/ root. Install requirements.txt, then run:

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python scripts/audit_gradient_horizon.py --evaluations
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The locked protocol contains fresh-run commands. Append --resume to the matching training command to recover a partial run. The artifact archive includes the parent, complete optimizer checkpoints, validation endpoints, inference exports, per-update histories, predictions, diagnostics and a SHA-256 manifest identifying the published source commit and tree.